

6th International Conference on Industry 4.0 and Smart Manufacturing

Advancing Task Allocation in Human-Robot Collaboration with a Multi-Simulation based Digital Twin System

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Abstract

The rapid advancement of Industry 4.0 has revolutionized manufacturing by integrating digital technologies to create smarter, interconnected systems. A key component within this shift is Human-Robot Collaboration (HRC), where humans and robots operate synergistically, combining human adaptability with robotic precision to enhance productivity and safety. Despite the potential of HRC, challenges persist in enabling real-time, operator-driven task allocation due to limitations in data synchronization and accessible interfaces. This research addresses these gaps by developing a multi-simulation-based Digital Twin (DT) system designed to manage task allocation in HRC settings. Integrating ergonomic and productivity key performance indicators (KPIs), the system empowers operators through a user-friendly interface for task management. A case study in automotive brake assembly showcases improvements in cycle times and ergonomic safety through proper task distribution.

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Peer-review under responsibility of the scientific committee of the 6th International Conference on Industry 4.0 and Smart Manufacturing

Keywords: Human-Robot Collaboration; Digital Twin; Operator 4.0; Ergonomics; Task Allocation;

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1. Introduction and State of the art

The advent of Industry 4.0 has transformed the manufacturing landscape by integrating advanced digital technologies to create smarter, more efficient, and interconnected production systems [1], [2]. This revolution has paved the way for new paradigms such as Human-Robot Collaboration (HRC), where human workers and robots synergistically operate within intelligent production environments to enhance operational efficiency and safety [3], [4]. HRC leverages the precision and endurance of robotic systems alongside the adaptability and problem-solving capabilities of humans, aiming to achieve higher levels of productivity and innovation [5] [6].

Central to the evolution of HRC is the integration of simulation technologies and Digital Twins (DTs), which enable the design, testing, and optimization of complex manufacturing processes in virtual environments [7], [8], [9]. Simulation provides a risk-free platform to model and analyze the behavior of manufacturing systems under various conditions without the costs associated with physical trials [10]. Building upon this, DTs offer dynamic, real-time representations of physical assets and processes, facilitating continuous monitoring and predictive analysis to support informed decision-making and rapid adaptability to changing production demands [11], [12].

Despite the advancements, the integration of DTs in HRC presents challenges such as data synchronization, accurate intention prediction, scalability, and communication latency [13]. Moreover, existing DT-based solutions often lack direct accessibility for line operators, rendering them off-line systems that limit real-time implementation of results [14], [15]. There is a pressing need for operator-centric tools that empower human workers to define constraints, express preferences, and evaluate proposed solutions in real-time, aligning with the principles of Industry 5.0, which emphasizes human creativity and strategic thinking within technologically advanced production environments [16], [17].

The concept of the Smart Operator (Operator 4.0) emerges as a forward-thinking approach within the manufacturing sector, embodying the integration of human intelligence with digital technologies [5], [18]. Operator 4.0 initiatives in HRC scenarios focus on empowering human workers to work alongside robots, leveraging both human flexibility and robotic precision to optimize operational efficiency and promote a safe working environment [19], [20]. However, a significant gap exists in the development of systems that enable direct operator engagement in task allocation, particularly in managing the trade-offs between productivity and ergonomics [21], [22].

This research aims to address these gaps by developing a multi-simulation-based Digital Twin system designed for dynamic task allocation in HRC contexts. The proposed system integrates both ergonomic and productivity KPIs, supporting data exchange and empowering operators through a user-friendly interface for dynamic task management. By leveraging established ergonomic methodologies and commercially available software, the system ensures reliability and reproducibility across various applications, aligning with the holistic approach necessitated by the fragmented nature of current research [23], [24].

The remainder of this paper consists of: a brief description of the applied methodology in Section 2; the case study presentation in Section 3; Section 4 will focus on the development of the said system; and finally in Section 5 a brief discussion of early results will be given.

2. Materials and Methods

The proposed research work is intended to develop a multi-simulation-based DT system that can be used by line operators for defining task allocation between humans and robots. Figure 1 shows the methodology that needs to be carried out to develop the overall system step by step [25]. The first 3 steps were covered in previous publications, while Step 4 and 5 are covered in this study:

- *STEP 1* - Identification of the system requirements and features.
- *STEP 2* - Definition of the system conceptual model.
- *STEP 3* - Design of the system architecture.
- **STEP 4** - Development of the system: this step transforms the conceptual designs and architectural plans into a working solution.
- **STEP 5** - Validation of the system: the final step involves assessing the effectiveness of the system in supporting line operators in managing task allocation.

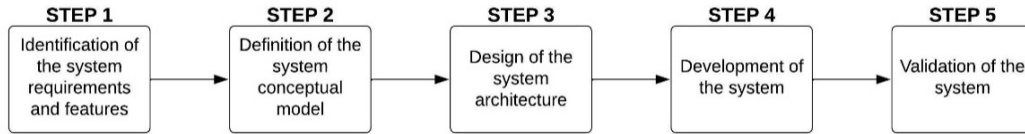


Figure 1. Step by step methodology

Figure 2, on the other hand, shows the outcome of the first three steps that is the system architecture.

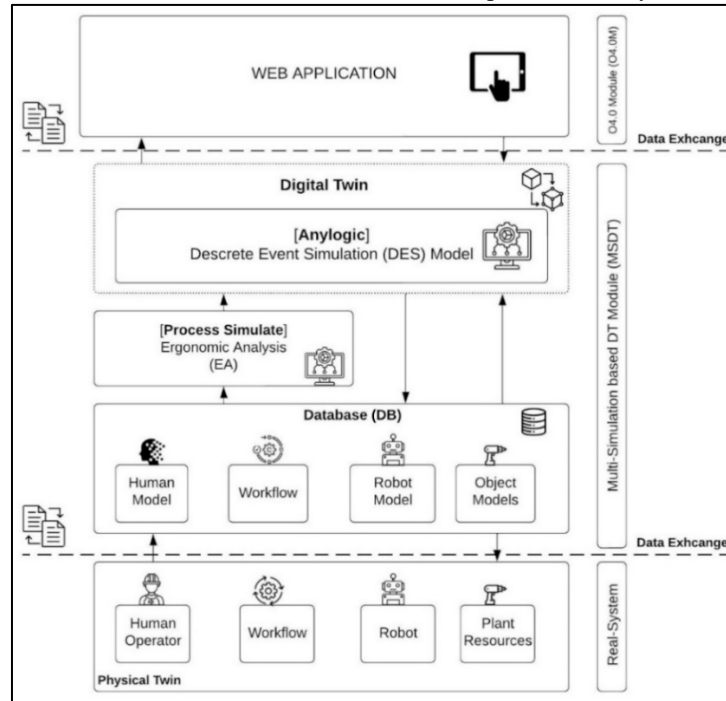


Figure 2. System architecture

The architecture consists of three modules: the Real System, the Multi-Simulation based Digital Twin Module (MSDTM) and the O4.0 Module (O4.0M). The MSDTM integrates an Ergonomic Analysis simulation model based on Process Simulate, a DES model based on Anylogic, and a database (DB). Input data are collected from the physical system, stored in the DB, and fed to the EA and DES modules. The DES module, which simulates assembly operations, assesses key performance indicators (KPIs), and produces task allocation scenarios. Users interact with these scenarios through the O4.0M, a graphical user interface (GUI) where they can accept, reject, or adjust the simulations. Data exchange between the physical and digital systems is facilitated by FIWARE's Orion Context Broker (CB), with Sender and Receiver applications managing data flow. A tablet-based user interface allows operators to provide real-time feedback. This setup enables synchronization, dynamic adaptation, and enhanced decision-making.

3. Case study

The proposed multi-simulation-based DT system aims to improve on-(the)-line task allocation within the HRC domain in assembly processes. The development of this system will be illustrated through a specific case study: an assembly workstation for a car tire brake system within the automotive industry. This workstation involves a human operator and a robot, currently in the implementation phase, that is supposed to assist in operational tasks. The robot's responsibilities include handling and delivering necessary tools and assembly parts or retrieving tools no longer in use. For the sake of clarity, since robot implementation is still a work-in-progress and thus some of the capabilities of the robot itself are under investigation, part of the operations that are still in evaluation here are considered as feasible

for collaboration activity. The complete list of assembly operations conducted at this workstation is provided in Table 1, along with information on the feasibility of potential robot support for each operation. It's worth noting that operations where HRC feasibility applies can be conducted either by the human operator alone or with the support of the robot. Each operation can be characterized by the following four different types of time: (i) Human time: this represents the average time required for the operator to perform the operation without robot support; (ii) HRC time: if HRC feasibility is applicable, this denotes the average time for collaboration between the human and the robot, such as when the robot assists the operator in handing over tools or components; (iii) Robot loading time: if HRC feasibility is applicable, this indicates the average time needed for the robot to prepare itself for collaboration with the operator, such as reaching and grasping a tool for assembly before handing it over to the operator. (iv) Robot unloading time: if HRC feasibility is applicable, this refers to the average time required for the robot to complete the operation after collaboration, such as returning a tool for assembly to its original position after use; (v) Approx. Weight: reports the weight of the components to be assembled. The time values presented in the table are derived from an analysis of historical data collected from the assembly workstation.

Table 1. List of workstation operations along with HRC feasibility, times and approximate weights

Op#	Description	HRC feasibility	Human time [sec]	HRC time [sec]	Robot loading time [sec]	Robot unloading time [sec]	Approx. Weight [kg]
1	Retrieve brake caliper	✓	30	20	5	3	3 kg
2	Position brake caliper	✓	20	15	3	2	(see above)
3	Install brake pads		40				1 kg per set
4	Retrieve brake rotor (disc)	✓	25	18	5	3	7 kg
5	Position brake rotor	✓	25	18	4	3	(see above)
6	Secure brake rotor with screws		35				~ 0 kg
7	Mount caliper assembly	✓	30	22	5	3	4 kg
8	Install caliper mounting bolts		40				~ 0 kg
9	Tighten caliper bolts		30				(see above)
10	Connect brake hose	✓	25	18	4	3	0.5 kg
11	Secure brake hose with clips		20				~ 0 kg
12	Retrieve master cylinder	✓	25	18	5	3	1.5 kg
13	Install master cylinder	✓	35	25	5	3	(see above)
14	Connect brake lines		50				0.3 kg total
15	Secure brake		20				(see above)
16	Fill brake system with brake fluid		40				1 kg (~ 1 L)
17	Bleed brake system	✓	80	60	10	5	(see above)
18	Inspect connections		30				~ 0 kg
19	Install wheel	✓	40	30	5	3	20 kg
20	Clean up workspace	✓	30	20	5	3	~ 0 kg

4. Development of the system

In this section, the authors discuss system development, explaining the interactions between modules and with the real world. Section 4.1 offers an overview of the data employed for system development, followed by discussions on the Process Simulate simulation model development in Section 4.2, the Anylogic DES module in Section 4.3. Finally, Section 4.4 summarizes details about the GUI.

4.1. System input data

The Process Layer serves as the foundational tier within the multi-layered architecture of the system. It contains the 'Real System / Physical Twin,' which includes the tangible components of the manufacturing environment: Human Operators, Plant Resources, Work Orders and Robots. Human Operators are the actual workers who interact with machines and carry out various tasks within the production line. Plant Resources refer to the machinery, equipment, and tools that are utilized in the manufacturing process. Work Orders are the directives that drive the manufacturing process, serving as detailed instructions for production tasks. They outline what, how, and when of production, guiding Human Operators and Robots in their respective functions. Robots represent automated systems designed to perform tasks with precision and efficiency, working in collaboration with human operators.

The information extracted from the Process Layer are fed to the Database within the Multi-Simulation layer fulfilling the data requirement from the simulation models. In particular, Table 2 summarizes the information taken from the different components of the Process Layer and fed into the Multi-Simulation Layer.

Table 2. Input Data Model

Module	Data	Used In
Human Operator	Operator ID	Anylogic, Process Simulate
	Anthropometric Parameters	Process Simulate
	Current Assigned Tasks	Anylogic
Plant Resource	Resource ID	Process Simulate
	CAD Model	Process Simulate
	Weight	Process Simulate
Work Order	Order ID	Anylogic
	Task Sequence	Anylogic, Process Simulate
	Task Execution	Process Simulate
	Task Characteristics	Anylogic, Process Simulate
	Current Task Allocation	Anylogic
Robot	Robot ID	Anylogic
	Current Assigned Tasks	Anylogic
	Task Allocation	Anylogic

The real-world data and interactions from the Process Layer are fed into the Database, which is part of the Multi-Simulation Layer. Within the Database, representations are created for the three types of entities: Human Model, Object Models, and Robot Model. The Human Model simulates the actions and interactions of human operators, the Object Models represent the various components and products being handled, and the Robot Model emulates the behavior and tasks of the robotic systems. With respect to the work orders, they are used in the form of Excel spreadsheets or Databases from which to extract necessary data for simulation.

The Multi-Simulation Layer operates above the Process Layer. This layer is tasked with the complex role of simulating and analyzing the operations and interactions that occur within the real system. It is comprised of the DES model running in Anylogic and addressing the “Process” side of the system simulation, and of the Process Simulate simulation model that is developed to address the human ergonomics side of the system.

4.2. Process Simulate

Process Simulate, a component of Siemens Tecnomatix software suite, is a robust and advanced simulation tool designed to facilitate the planning, testing, and optimization of manufacturing processes within the digital environment. In order to develop the simulation model on Process Simulate, the first step is the collection of anthropometric data to describe the physical characteristics of the operators involved in the assembly line. Process Simulate by means of its module relative to the Human Modelling (Jack&Jill), offers the possibility to set up specific anatomic parameters. After collecting human data, then it is necessary to collect resource data: the conversion of CAD

models (using the CAD Translator feature) into .jt format, uploading them to the simulation scenario, and assigning weight properties to the components used by the operator. The design of the workstation layout is performed together with the study of the assembly procedures through time measurements and video reviews to ensure that the operations can be digitally recreated with as much accuracy as possible.

In order to recreate the assembly procedures as closely as possible to the originals, the function of Process Simulate called Task Simulation Builder (TSB) is used. What is obtained from the digital replication of operations carried out in the real environment is a diagram of sequential operations. After replicating the real process within the digital environment, it is time to conduct ergonomic analyses. Process Simulate offers some of the most recognized methodologies to conduct ergonomic assessments, including: Owako Working-posture Analyzing System (OWAS) [26]; The National Institute for Occupational Safety and Health (NIOSH) [27]; European Assembly Work-Sheet (EAWS) [28]; Low Back Analysis (LBA) [29]; Energy Expenditure [30].

4.3. Anylogic

As previously mentioned, the ergonomic module related to Process Simulate provides important ergonomic report data as input to the Anylogic model, thus supplying information for defining the task allocation between humans and robots. The solution developed in Anylogic (version 8) is showed in Figure 11.

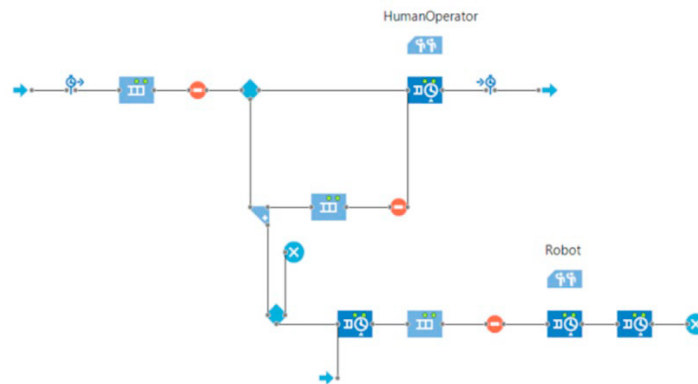


Figure 3. Model flow chart developed in Anylogic 8

The idea behind the model is to create a loop in which each assembly operation (task) is represented by an agent that moves through the system as many times as necessary to complete whole cycle. Indeed, the agent represents the dedicated task in each cycle. This approach offers several advantages: (i) all the information needed to model the process can be easily entered into one (or more) Excel sheets (or other types of databases); (ii) the simulation model is thus completely generalized for the various possible configuration options in collaborative contexts; (iii) from a programming effectiveness perspective, the simulation model will be easily scalable and flexible, since if one wants to intervene on the process logics, it will suffice to simply change the information provided as input with almost none or no effort required to readjust the simulation model; (iv) the possibility to manage the simulation model solely through a dedicated user interface for the line operator and to handle the input information directly on the database sheets, thus isolating the core of the simulation system and preventing users without the appropriate skills from intervening directly on the model; (v) it will be possible to easily reuse the same system architecture should there be a need to implement this type of simulation model in another case study context or another workstation within the same production system. Data within the simulation model are read from databases (in this case study, Excel sheets) through specific programming commands placed in the appropriate simulation blocks. Within the simulation model, through programming codes, four logics are considered: (i) the current operation performed only by the operator and the next operation to be performed only by the operator, (ii) the current operation performed collaboratively and the next operation in collaboration, (iii) the current operation performed only by the operator and the next operation to be performed in collaboration, (iv) the current operation performed in collaboration and the next operation to be performed only by the operator. Cases (i) and (ii) are obvious from a simulation standpoint. As for case (iii), according

to the implemented logic (using Boolean variables and checkpoints), the robot is able to prepare to support the operator by taking over the next task and performing the loading operation while the operator is busy with the current task (thus promoting a lower cycle time). Similarly, regarding case (iv), the operator within the simulation is able to continue to the next operation to be performed independently while the robot carries out the unloading phase of the previous operation. The main settings for the DES are set by the operator through a dedicated Graphic User Interface (GUI) installed on a device directly on the line.

4.4. Graphic User Interface (GUI)

The O4.0 layer is defined as such because it extends and enhances the capabilities of the line operator. This enhancement is achieved because the tool directly supports the operator involved in the production or assembly line. It empowers operators to make more judicious and informed decisions by referencing KPIs that they would otherwise be unable to evaluate by himself/herself. This advanced support mechanism facilitates real-time decision-making in production processes, ensuring operators can respond quickly and effectively to changing conditions and requests. The entire system is interfaced through a Web Application with a GUI. The GUI is available on a tablet installed on the human operator workstation. The user interface is designed to be user-friendly, intuitive and easy to use. The GUI has to guarantee that the operator can set-up a simulation and run it in few moments and without any time waste. For this reason, the minimal design of the interface has been preferred. Figure 12 shows the first screen that operator faces when accessing the tablet.

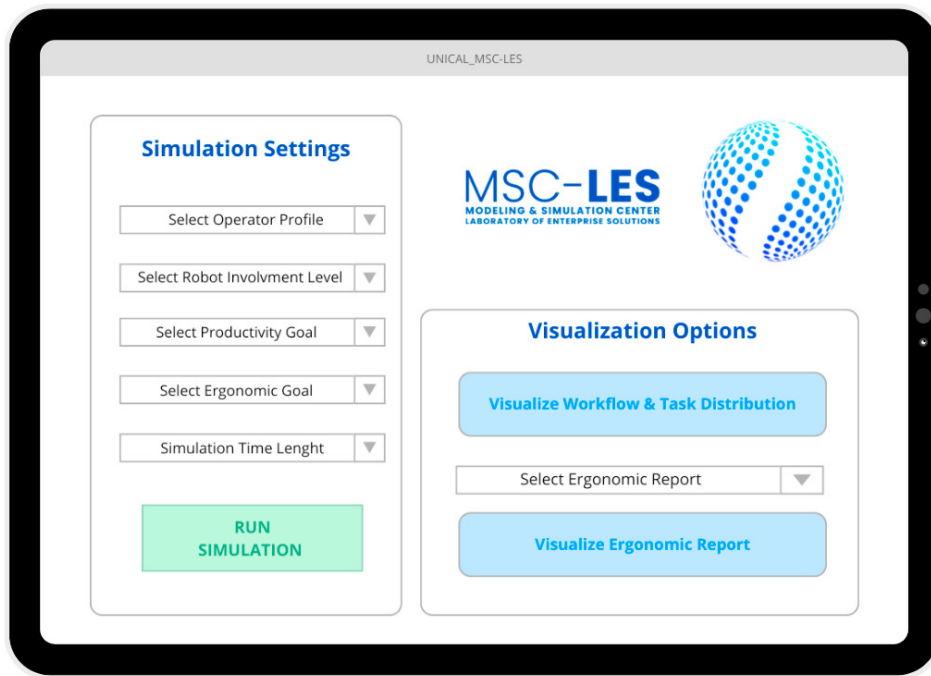


Figure 4. GUI Home Page

From the Home Page the line operator has different possibilities: set-up the simulation settings or visualize workflow, task distribution or the ergonomic report. Among the simulations settings, the operator can set up: (i) Operator Profile. Each operator that operates in the specific workstation has a profile in which the system stores data useful for ergonomic analysis. (ii) Robot Involvement Level. The robot can be involved in the production with three different rates: 0% the robot does not take any operation; 50% the robot is potentially involved in taking only half of the possible operation for which it is able to intervene; 100% the robot is potentially involved in all of the tasks that it can take into account. (iii) Productivity goal. The operator can select the goal function that he intends to address for task allocation among: Cycle Time, Throughput, Tardiness, Human Operator/Robot Utilization Levels. (iv)

Ergonomic goal. The operator can select the goal function that he intends to address for task allocation among: Back Load, Arms Load, Leg Load, Neck Load, Energy Expenditure. (v) Simulation Time. The operator can decide the length of the simulation timespan (1 hour, 3 hours, 5 hours).

On the other hand, the operator can choose to visualize the workflow and task distribution or the ergonomic report. Once the operator has pressed the button to run the simulation, and the simulation has been completed, then the operator can choose to visualize the workflow and new task distribution or the simulation insights. Based on the insight's obtained, then the operator can decide to accept the new task allocation or to decline and try to perform another simulation with other parameters. Once the operator has accepted the new task allocation, then the operator can see from the tablet the current task he has to execute and if the specific task is carried out in collaboration or not.

5. Results and Discussion

In order to validate the multi-simulation-based DT, in this section, we are going to consider the case in which the human operator has set-upped the simulation parameters by choosing his profile (let's say *Jack*), a robot involvement level of 100%, which means that the robot can take up to the 100% of all the task in its capacity, picked the reduction of the Cycle Time as productivity goal. The simulation time length has been selected as *4 hours*. The line operator can now run the simulation, producing the insights useful to evaluate the system proposed task allocation. Figure 5 (on the left) shows the Cycle Time Chart, y-axis indicates the amount of time, expressed in seconds, taken to complete task while the x-axis shows tasks numbered from 0 to 20. Three different functions are reported on the chart: (i) Human-only (green), represents the time taken for tasks when completed individually by a human without assistance. (ii) Full Robot Employment (orange): depicts the time taken for tasks when the robot intervenes in all the tasks within its capabilities. (iii) Fastest Task Allocation (blue) represents a scenario where tasks are allocated between humans and robots considering objective functions.

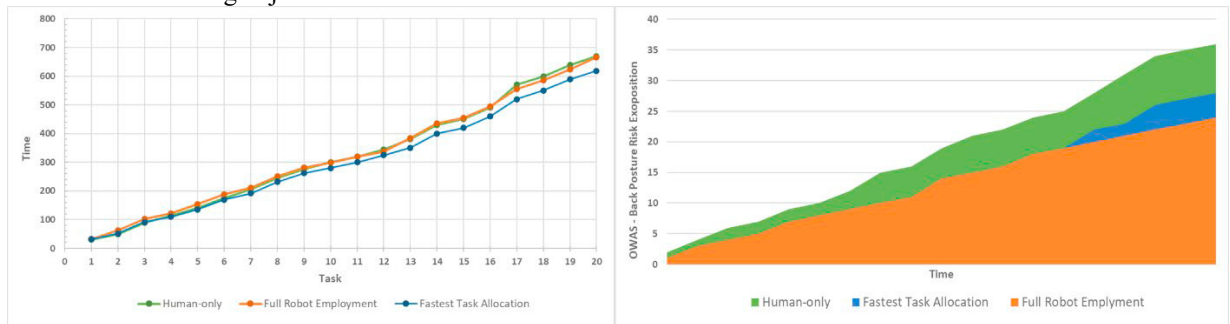


Figure 5. Cumulative Cycle Time (on the right) and cumulative Back posture Risk Exposition

The proposed chart on the left shows that, in terms of cycle time reduction, there is minimal difference (-0.75%) between the Human-only and Full Robot modes. This is because, in the Full Robot configuration, the operator often must wait for the robot's Load and Unload procedures between consecutive collaborative tasks. This observation is supported by the data in Figure 6 (left side), where the Human Mean Utilization Level (indicating the ratio of time the operator spends on assembly tasks versus waiting) is around 80%. In contrast, under the cycle time minimization configuration, this value reaches around 90%. The Human-only case is not shown in the graphs in Figure 6, as it would naturally result in a Human Mean Utilization Level of 1 and a Robot Mean Utilization Level of 0. Furthermore, from Figure 5, the comparison between the blue curve (Fastest Task Allocation) and the green curve (Human-only) indicates that this specific task allocation leads to a cycle time improvement of -7.61%. Additionally, as shown in Figure 6 (right side), the robot is not involved in all possible tasks under the Fastest Task Allocation configuration but is selectively deployed for tasks with greater strategic impact. This is evident as the Robot Mean Utilization Level is lower in the Fastest Task Allocation configuration.

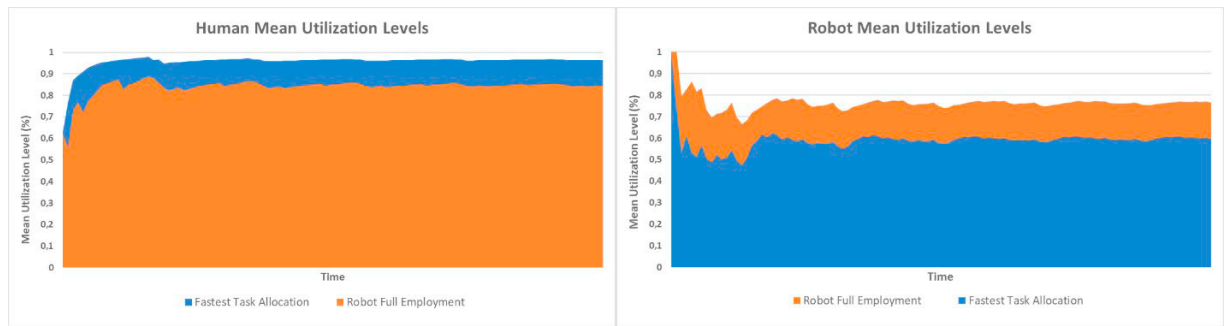


Figure 6. Comparison between Mean Utilization Levels in Robot Full Employment Scenario and Fastest Task Allocation Scenario: (left) Human Operator, (right) Robot

One final aspect worth highlighting is the chart on the right side of Figure 5, which shows the cumulative trend of Back Posture Risk Exposure computed through OWAS analysis. What emerges is that, compared to the baseline represented by the Human-only configuration, assigning all possible tasks to the robot results in a substantial reduction in risk exposure (-33%). The Fastest Task Allocation configuration also yields a meaningful, though more modest, reduction in risk exposure (-22%).

6. Conclusion

This research presents a multi-simulation-based Digital Twin system designed to enhance task allocation in HRC settings. By integrating ergonomic and productivity KPIs, the system empowers line operators through a user-friendly interface, aligning with the principles of Operator 4.0 and Industry 5.0. The case study on automotive brake assembly demonstrated significant improvements in cycle time and ergonomic risk exposure. Specifically, the “fastest task allocation” scenario reduced the cycle time by 7.61% compared to the human-only configuration and decreased ergonomic risk exposure by 22%. These results validate the effectiveness of the proposed system in enhancing both productivity and worker well-being.

Furthermore, the system's dynamic task allocation capability represents a significant advancement in HRC environments. By allowing quick adjustments based on operator input and system feedback, the Digital Twin facilitates a more responsive and flexible production process. This adaptability not only improves efficiency but also enhances the ergonomic conditions for workers by distributing tasks in a manner that mitigates physical strain. The integration of ergonomic assessments into the task allocation algorithm ensures that worker health is prioritized alongside productivity goals, embodying a holistic approach to manufacturing system design.

Future work will focus on further refining the system by incorporating machine learning algorithms to enhance predictive capabilities. Additionally, expanding the system's applicability to other manufacturing contexts and integrating more advanced ergonomic assessment tools will be valuable. The ongoing development of operator-centric tools within HRC environments holds the potential to revolutionize manufacturing processes by fostering greater adaptability, efficiency, and human-centric design, fully embracing the ideals of Industry 5.0.

Acknowledgements

This research work was partly supported by the FELICE Project – Flexible Assembly Manufacturing with Human-Robot Collaboration and Digital Twin Models – funded from the European Union's Horizon 2020 Research and Innovation program under grant agreement No. 101017151.

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